Kunal Mathur

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GitHub: github.com/kunal202426 | Portfolio: portfolio-kunal26.vercel.app

SUMMARY

Software engineer (B.Tech CSE, VIT) and founding engineer at Astra, building full-stack product features on Next.js 16, React 19 and Supabase/PostgreSQL. Former intern on YES Bank's OMNI trading platform. Works across TypeScript, React, REST APIs and Node.js, with projects in real-time Kafka and Redis reconciliation (Java, Spring Boot), distributed-systems failure simulation and AI agent tooling. IIT Madras Global Hyperloop DesignX winner; Flipkart GRiD 7.0 national semifinalist.

EXPERIENCE

Astra Sep 2026 – Present
Founding Engineer *Bengaluru, India (On-site)*

- Developed full-stack product features across job and internship discovery, campus placement prep, question banks, AI-powered resume, GitHub and portfolio analysers and a company map, on Next.js 16 (App Router), React 19 and TypeScript with Supabase (PostgreSQL, Auth, Row-Level Security), deployed on Vercel.
- Delivered the recruiter and candidate Talent workspace end to end (**700+ commits** across the Astra codebase) on a multi-tenant Postgres model: wrote **114 SQL migrations**, **57 rollback scripts** and **46 database tests**, added consent-gated messaging, scoring and a leaderboard, and removed redundant fetches; also created the Commercial staff admin panel (billing, usage, offline payments, role-based access, action log).
- Created Astra Agent, a Chrome Manifest V3 extension (TypeScript, Vite) that autofills Greenhouse, Lever and Ashby applications from a Supabase profile API under RLS; added a job-ingestion pipeline (Chrome DevTools Protocol source, parser, URL normalization, promotion classification) and Astra Wbot, a WhatsApp jobs bot (TypeScript, SQLite, Docker Compose, Vitest) with a daily digest and admin dashboard.

YES Securities (YES Bank) Jan 2026 – Sep 2026
Full Stack Developer Intern *Bengaluru, India*

- Shipped **5+ production features** (React, Node.js, REST APIs) on the OMNI trading platform serving **20,000+ daily trades**, working in Agile sprints with code reviews; refactored backend microservices, cutting average API latency **~25%**.
- Diagnosed N+1 query bottlenecks and redesigned database schemas, cutting query time **~35%**; streamlined CI/CD across 3 backend microservices.

PROJECTS

CodeGraph: GraphRAG Memory Layer for AI Coding Agents | GitHub 2026 – Sep 2026

- Designed a local-first memory layer for AI coding agents: DuckDB entity/edge graph, 22-language parsing (tree-sitter), and embeddings fused with literal search via Reciprocal Rank Fusion, exposed as a 12-tool MCP API; **58% cheaper** on deep impact-analysis questions in a real A/B test on a 97,000-entity repo.

Hybrid Stock Price Prediction System | GitHub | Live Jan 2026 – Apr 2026

- Implemented an end-to-end forecasting app: ETL pipelines ingest OHLCV market and news data into a FastAPI REST layer that feeds a React dashboard; an LSTM, LightGBM and FinBERT ensemble cut MAE **~18%** versus single-model baselines.

EDUCATION

Vellore Institute of Technology 2022 – 2026
B.Tech, Computer Science & Engineering *Vellore, Tamil Nadu, India*

ACHIEVEMENTS

Winner: Global Hyperloop Competition (Asia's First), IIT Madras 2025

- Led software development as AI & Software Lead of Team Levitate (VIT), a cross-functional team, under competition deadlines; won the DesignX trophy and **5 category awards** against 15+ international teams.

National Semifinalist: Flipkart GRiD 7.0 2025

- Advanced to the national semi-finals of Flipkart GRiD 7.0, a national competition with timed algorithmic coding rounds.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL
CS Fundamentals: Data Structures and Algorithms (DSA), OOP, Operating Systems, DBMS, Computer Networks, Multithreading, Performance Optimization
Frontend: Next.js, React, Redux, Tailwind CSS, shadcn/ui, Three.js, Vite, HTML, CSS
Backend & Data: Node.js, Express.js, Spring Boot, FastAPI, REST APIs, GraphQL, gRPC, Microservices, Distributed Systems, WebSockets, Webhooks, Kafka, PostgreSQL, MySQL, MongoDB, SQLite, Supabase, Redis, Caching, Schema Design, System Design, RBAC, Cron Jobs
Tools: Git, CI/CD, Agile, Docker, Docker Compose, Kubernetes, Vitest, pytest, Vercel, AWS, Debugging, Monitoring
AI: LLM APIs (Claude, Gemini), MCP, Embeddings, Prompt Engineering, Machine Learning